“Exploring the Impact of Natural Environments on Birth Outcomes: A Research Proposal Exercise”

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Abstract: (1) Background: Maternal exposure to natural environments, such as green and blue spaces, has been increasingly recognized for its potential to improve pregnancy outcomes. However, evidence remains mixed, particularly regarding the effects of blue spaces on birth outcomes. This study investigates the relationship between maternal exposure to green and blue spaces and two critical pregnancy outcomes: birth weight and preterm birth. The aim is to provide actionable insights to inform public health and urban planning policies. (2) Methods: This study analyzed data from 4,445 participants, incorporating maternal, environmental, and birth outcome variables. Birth weight was modeled as a continuous variable using linear regression, while preterm birth was examined as a binary outcome using logistic regression. Predictors included proximity to green and blue spaces, accessibility to these environments, air pollutant levels (NO_2 and $PM_{2.5}$), and maternal socioeconomic factors. Cohort-level variability was addressed using Generalized Linear Mixed Models (GLMMs). Multicollinearity, residual diagnostics, and model robustness were systematically assessed. (3) Results: Proximity to green spaces was significantly associated with increased birth weight, with each unit increase in distance corresponding to a reduction in birth weight. Blue spaces showed a notable, though less consistent, protective effect against preterm birth. Elevated NO_2 and $PM_{2.5}$ levels were linked to decreased birth weight and increased preterm birth risk. Cohort-level variability highlighted significant local disparities, emphasizing the influence of community-level factors on pregnancy outcomes. (4) Conclusions: These findings underscore the critical role of natural environments and air quality in shaping pregnancy outcomes. Proximity to green and blue spaces and lower air pollution exposure were associated with better maternal and fetal health. Urban planning and public health interventions that enhance access to natural spaces and reduce air pollution can substantially improve birth outcomes, particularly in high-risk populations.

1. Introduction

Maternal exposure to green and blue spaces has gained attention for its potential to improve pregnancy outcomes, such as increasing birth weight and reducing risks of low birth weight (LBW) and preterm birth (PTB) [1,2].
Existing evidence suggests that green spaces, like parks and other vegetated areas, promote maternal and fetal health through mechanisms such as stress reduction, improved air quality, and social support [3–5]. However, findings are mixed, particularly regarding the effects of blue spaces (e.g., proximity to water bodies) on birth outcomes, with limited data and inconsistent associations observed across studies [6,7].
With rising urbanization, the accessibility of natural spaces is declining, which could negatively impact fetal growth and birth outcomes among pregnant women living in urban areas. Understanding these associations is essential for guiding health recommendations for expectant mothers and for shaping urban planning policies that promote health equity. Studies consistently report positive associations between green space exposure and higher birth weight, although data on blue spaces are scarcer and often inconclusive [2,8]. Identifying the extent to which natural environmental exposure affects birth outcomes can offer insights for public health interventions aimed at enhancing access to green and blue spaces. Our study aims to investigate the influence of maternal exposure to green and blue spaces on fetal growth and birth weight. Specifically, we will analyze the relationship between green and blue space exposure and two key outcomes: (1) birth weight (as a continuous outcome analyzed through regression), and (2) risk of prematurity (as a binary classification outcome).

2. Materials and Methods

2.1 Study Population and Data Description

This study analyzed data from a total of 4,445 participants, including both categorical and numerical variables, with no missing values, ensuring completeness and robustness for all analyses. The dataset encompasses a wide range of maternal, environmental, and birth outcome variables to investigate the potential impact of exposure to natural environments on birth outcomes. The variables are summarized as follows:

- Maternal and Socioeconomic Variables: Mother's age (year), Parity (1 = multipara, 0 = primipara), Maternal education (High, Medium, Low), Maternal ethnicity (Western or non-Western), Residential area deprivation level (High, Medium, Low), Maternal smoking (Yes or No), Maternal alcohol consumption (Yes or No).
- Environmental Exposure Variables: Normalized Difference Vegetation Index (NDVI) within a 500/300/100-meter radius of the mother's residence, Residential distance to green space (meters), Accessibility to green space (Yes or No), Residential distance to blue space (meters), Accessibility to blue space (Yes or No).
- Air Pollution Variables: level of nitrogen dioxide ($\mu g/m^3$), Level of particulate matter $PM_{2.5}$ ($\mu g/m^3$).
- Birth Outcome Variables: newborn weight (grams), Preterm birth (1 = preterm, 0 = term).
- Cohort Variable: categorical variable for the study group, controlling for cohort-specific effects.

The variables used in this study were catalogued as follows:

- Categorical Variables: Parity, Preterm, Child sex, Maternal education, Maternal ethnicity, Area level deprivation, Maternal smoking, Maternal alcohol consumption, Accessibility to green space, Accessibility to blue space, Cohort.
- Numerical Variables: Maternal age, NDVI 100m, NDVI 300m, NDVI 500m, Residential distance to green space, Residential distance to blue space, NO_2, $PM_{2.5}$, Birth weight.

2.2 Descriptive Statistics

2.2.1 Analysis of Categorical Variables

Relative and absolute frequencies were calculated for categorical variables to summarize the distribution within each category.

2.2.2 Analysis of Numerical Variables

For numerical variables, descriptive statistics were calculated, including:

- Mean and Standard Deviation (SD): to represent central tendency and variability.
- Median and Interquartile Range (IQR): to capture robust measures of central tendency and spread, particularly for skewed distributions.

2.3 Statistical Testing

The primary outcomes were Birth Weight (numerical) and Preterm Birth (binary).
Associations between predictors and outcomes were tested as follows:

- Categorical Variables vs. Birth Weight: T-tests, to compare means for binary categorical variables, and ANOVA, used for multi-class categorical variables, such as Maternal education, Area level deprivation, and Cohort.
- Categorical Variables vs. Preterm: Chi-squared Tests to assess the association between categorical variables and Preterm status.
- Numerical Variables vs. Birth Weight: Pearson Correlation to measure the linear correlation between numerical variables and Birth Weight.
- Numerical Variables vs. Preterm: T-tests to compare mean values of numerical variables between Preterm and Non-Preterm groups.

P-values < 0.05 were considered statistically significant, and p-values $< 10^{-3}$ indicated very strong associations.

T-tests and ANOVA were applied even when data did not meet the normality assumption, as the sample size for each group was greater than 30, leveraging the Central Limit Theorem to justify the analysis.

2.4 Analysis of Covariates

To ensure accurate interpretation of the relationships between environmental exposures and birth outcomes, the evaluation of key covariates was focused on two main analyses:

1. Correlation Matrix for Numerical Variables: numerical covariates were assessed using a correlation matrix. This approach provided an overview of linear relationships between predictors and birth outcomes, highlighting potential multicollinearity or direct associations.

2. Variance Inflation Factors (VIF) for Multicollinearity Assessment: multicollinearity among all predictors, including numerical and categorical variables, was evaluated using Variance Inflation Factors (VIF). Predictors with a VIF >10 were considered highly collinear and were either excluded or transformed as necessary to maintain model stability and interpretability.

These methods ensured efficient evaluation of covariates, focusing on their relevance and appropriateness while minimizing redundancy in analyses. Polynomial terms were introduced for variables showing significant associations, enabling the modeling of potential non-linear relationships.

2.5 Regression Models

2.5.1 Linear Regression for Birth Weight

1. Data Preparation: the variables were chosen to comprehensively capture maternal characteristics, environmental exposures, and key birth outcomes, while ensuring no direct overlap between outcome predictors (e.g., Birth weight excluded as input for Preterm model). Predictors with significant associations ($p < 0.05$) were included in initial models. Predictors were categorized into:

- Numerical predictors, which were standardized using StandardScaler.
- Categorical predictors, transformed using One-Hot Encoding.

1. Model Development

- Initial Models: models were built using numerical predictors alone and extended to include polynomial interactions and categorical variables.
- Feature Selection: Best Subset Selection was applied, optimizing adjusted R^2, AIC, and BIC to identify the most predictive features.
- Cross-Validation: a 10-fold cross-validation strategy assessed model robustness and generalizability.
- Clustering: the dataset was clustered by Cohort to account for intra-group variations.
- Metrics such as R^2, adjusted R^2, and Mean Squared Error (MSE) evaluated model performance.

2. Assumptions and Diagnostics

- Independence: assessed using the Durbin-Watson test for residual autocorrelation.
- Gaussian Distribution: residuals were tested using Shapiro-Wilk tests and visualized via QQ plots.
- Linearity and Homoscedasticity: residuals vs. fitted values plots were analyzed for non-random patterns, and the Breusch-Pagan test was applied.
- Outlier and Leverage Analysis: influential points were identified and removed if they distorted model coefficients.

2.5.2 Logistic Regression for Preterm

1. Data Preparation: predictors significantly associated with Preterm status were included in logistic regression models. Data was standardized and One-Hot Encoded as necessary.

2. Model Development

- Threshold Selection: the classification threshold was set at 1−prevalence of Preterm, balancing sensitivity and specificity.
- Feature Selection: Best Subset Selection identified optimal predictors using adjusted R^2 and cross-validation.
- Cross-Validation: a 10-fold cross-validation strategy assessed model robustness and generalizability.
- Clustering: the dataset was clustered by Cohort to account for intra-group variations.

3. Performance Metrics

- ROC Curve and AUC: to assess discriminatory power.
- Precision, Sensitivity, Specificity, F1-Score, and Accuracy: to balance false positives and false negatives.

2.6 Generalized Linear Mixed Models

To account for clustering by Cohort, Generalized Linear Mixed Model (GLMM) were developed. Predictions included fixed and random effects for enhanced accuracy. Random intercepts were used for each cohort to account for within-group dependencies and were visualized using bar plots to interpret cohort-level variability.

2.7 Statistical Software and Visualization

All analyses were performed using Python:

- Data Manipulation: pandas and numpy.
- Statistical Tests: scipy.stats and statsmodels.
- Model Development: statsmodels for regression models, gpboost for GLMMs.
- Feature Selection: custom implementation of Best Subset Selection.
- Visualization: matplotlib and seaborn for histograms, heatmaps, scatterplots, and ROC curves.

3. Results

3.1. Descriptive Statistics

Among the 4,445 participants, 53% of mothers were primiparous, and 47% were multiparous. A significant proportion of births were preterm (40%). Maternal education levels were predominantly high (52%), followed by low (30%) and medium (18%). Western ethnicity constituted 78% of the sample. Area-level deprivation was categorized as high for 43% of participants, low for 39%, and medium for 18%. Notably, 82% of mothers reported no smoking during pregnancy, while 51% consumed alcohol. Regarding environmental exposure, 74% of participants had accessibility to green spaces, but only 21% had accessibility to blue spaces. Table 1 summarizes the distribution of key categorical variables.

Table 1 Distribution of categorical variables.

Variable	Categories	Absolute Frequency	Relative Frequency (%)
Parity	Single, Multi	2373, 2072	53, 47
Preterm	No, Yes	2646, 1799	60, 40
Child Sex	Male, Female	2229, 2216	50, 50
Maternal Education	High, Low, Medium	2308, 1345, 792	52, 30, 18
Maternal Ethnicity	Western, Non-Western	3476, 969	78, 22
Area Level Deprivation	High, Low, Medium	1927, 1736, 782	43, 39, 18
Maternal Smoking	No, Yes	3646, 799	82, 18
Maternal Alcohol Consumption	Yes, No	2270, 2175	51, 49
Accessibility to Green Space	Yes, No	3302, 1143	74, 26
Accessibility to Blue Space	No, Yes	3494, 951	79, 21
Cohort	J, C, B, D, I, F, A, H, G, E, K	1072, 658, 562, 527, 445, 409, 359, 202, 98, 70, 43	24, 15, 13, 12, 10, 9, 8, 5, 2, 2, 1

Table 2 Summary statistics for numerical variables.

Variable	Mean (SD)	Median (IQR)
Maternal Age (years)	29.68 (4.82)	29.86 (26.66–32.92)
NDVI 100m	0.42 (0.14)	0.41 (0.32–0.51)
NDVI 300m	0.43 (0.13)	0.43 (0.35–0.51)
NDVI 500m	0.44 (0.13)	0.44 (0.37–0.53)
Residential Distance to Green Space (meters)	220.60 (149.49)	194.09 (104.86–307.01)
Residential Distance to Blue Space (meters)	1096.83 (855.77)	943.98 (408.14–1539.52)
NO_2 (μg/m^3)	26.00 (9.88)	25.29 (19.36–31.80)
$PM_{2.5}$ (μg/m^3)	14.96 (4.65)	13.93 (11.38–18.09)
Birth Weight (grams)	2964.99 (872.96)	3050.16 (2367.79–3605.95)

The mean maternal age was 29.68 years (SD 4.82), with a median of 29.86 years. Among environmental exposure metrics, the NDVI scores were highest within a 500m radius (mean 0.44, SD 0.13). The mean residential distances to green and blue spaces were 220.60m and 1,096.83m, respectively. Air pollution levels showed a mean NO_2 concentration of 26.00 μg/m^3 (SD 9.88) and $PM_{2.5}$ concentration of 14.96 μg/m^3 (SD 4.65). Birth weight had a mean of 2,964.99g (SD 872.96), with a median of 3,050.16g. Table 2 provides detailed summary statistics for numerical variables.

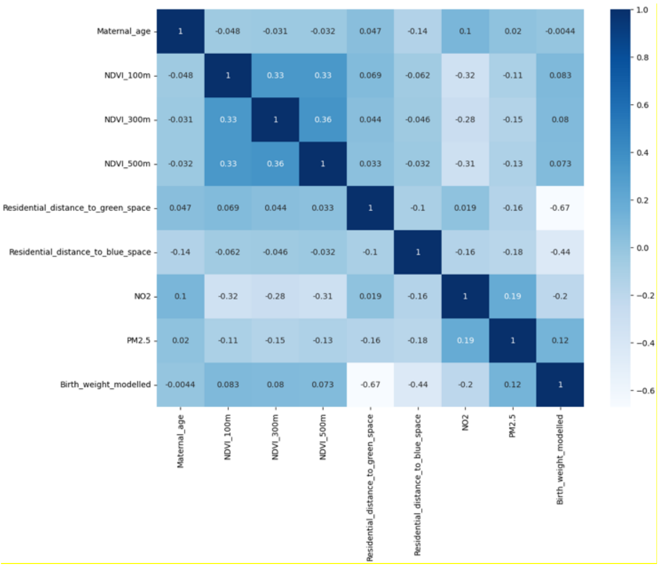
3.2. Covariance Analysis and Multicollinearity Assessment

Understanding the relationships between predictors is a critical step in building robust predictive models. This analysis aims to identify associations among variables, assess potential multicollinearity, and make informed decisions on variable inclusion in the final model. Two key techniques were employed: the correlation matrix and the Variance Inflation Factor (VIF).

3.2.1 Correlation Matrix

The correlation matrix (Figure 1) offers a visual representation of the linear relationships between numerical predictors. It provides insights into potential associations that may influence variable inclusion in the regression model.

Figure 1 Correlation matrix among the numerical variables.



Key Observations:

1. NDVI Metrics (100m, 300m, 500m)
   The NDVI metrics exhibit strong intercorrelation (r = 0.33–0.36), indicating consistent vegetation coverage across spatial radii. This redundancy could lead to multicollinearity in the model, complicating the isolation of vegetation's independent effects on outcomes. To address this, composite indices should be considered to streamline the model while preserving essential environmental information.
2. Residential Distance to Green and Blue Spaces

- Green Space: A strong negative correlation with birth weight (r = -0.67) suggests that proximity to green spaces is associated with better fetal growth. This aligns with evidence showing that green spaces reduce stress and improve air quality.
- Blue Space: A moderate negative correlation with birth weight (r = -0.44) supports the idea that access to water bodies can benefit maternal health, though its effect is weaker compared to green spaces.
  Proximity to green and blue spaces stands out as critical predictors of birth outcomes and should be prioritized.

3. Air Pollution (NO_2 and $PM_{2.5}$)

- NO_2 and NDVI Metrics: Negative correlations (r = -0.28 to -0.32) indicate that greener areas tend to have lower NO_2 levels, reinforcing the environmental benefits of vegetation.
- Both NO_2 and $PM_{2.5}$ show weak correlations with birth weight (r = -0.20 and r = 0.12, respectively), indicating that their impact might be more indirect and context-dependent.

3.2.2 Variance Inflation Factor (VIF)

1. Initial VIF Results:

- Maternal Age (VIF = 38.61): a very high VIF indicates strong collinearity with other predictors. This redundancy justifies its exclusion from the model.
- NDVI Metrics (VIF = 15.43–18.73): the high VIF values reflect the correlations among NDVI_100m, NDVI_300m, and NDVI_500m, confirming the need for a composite index (NDVI_100m × NDVI_300m × NDVI_500m).
- Air Pollution ($PM_{2.5}$ = 52.24, NO_2 = 13.81): the high VIF values for $PM_{2.5}$ and NO_2 indicate strong multicollinearity. A composite variable (e.g., $NO_2 \times PM_{2.5}$) can capture their combined effects without introducing redundancy.
- Residential Distance Variables: moderate VIF values for residential distance to green space (7.15) and blue space (6.78) indicate acceptable levels of multicollinearity, allowing them to remain in the model.

2. Refined VIF Results:

After excluding redundant variables and introducing composite indices ($NO_2 \times PM_{2.5}$ and NDVI_100 × NDVI_300 × NDVI_500), the highest VIF dropped to 8.99. Key changes include:

- $NO_2 \times PM_{2.5}$ (VIF = 8.99): captures the joint effect of air pollution metrics while reducing multicollinearity.
- NDVI Composite (VIF = 6.78): simplifies the representation of vegetation indices across radii.
- Residential Distances: moderate VIF values (green space = 3.63, blue space = 5.64) confirm their independence.

The refined VIF results (Table 3) confirm that multicollinearity has been effectively addressed, enhancing the reliability and interpretability of the model.

Table 3 Final VIF Values After Variable Refinement and Composite Index Introduction to Address Multicollinearity

Variable	VIF
Residential distance to green space	3.63
Residential distance to blue space	5.64
$NO_2 \times PM_{2.5}$	8.99
NDVI composite	6.78
Preterm	3.09
Parity	2.16
Child Sex	1.96
Maternal Education (Low, Medium)	1.83–1.88
Maternal ethnicity	5.58
Area Deprivation Level (Low, Medium)	2.10-1.62
Maternal smoking	1.26
Maternal alcohol consumption	2.45
Accessibility to green space	3.81
Accessibility to blue space	2.12
Cohorts	1.38–5.09

3.3. Statistical Testing

The results presented in Table 4 highlight key associations between various maternal, environmental, and socioeconomic factors and the outcomes of birth weight and preterm delivery. Among maternal characteristics, ethnicity and alcohol consumption showed significant associations with both birth weight and preterm delivery, suggesting that these factors may play an important role in shaping birth outcomes. In contrast, parity, child sex, maternal education, and smoking were not significantly associated with either outcome, indicating that their effects may be less pronounced or confounded by other variables in this dataset.

Environmental exposures, particularly proximity to natural spaces, emerged as strong predictors. Accessibility to green and blue spaces was consistently associated with higher birth weight and a lower likelihood of preterm delivery. These

findings were further supported by the NDVI metrics, which showed significant positive associations with both outcomes across different spatial scales.
Residential distances to green and blue spaces were also strongly negatively associated with both birth weight and preterm delivery, reinforcing the idea that closer proximity to natural environments is beneficial.
Air pollution, as measured by NO_2 and $PM_{2.5}$ levels, was found to negatively impact both outcomes. While NO_2 primarily affected birth weight, $PM_{2.5}$ was significantly associated with both birth weight and preterm delivery, underscoring the detrimental role of air pollutants during pregnancy.
Finally, cohort-level differences were highly significant, reflecting variations in outcomes across different study groups. This justifies the use of cohort clustering in the analysis to account for group-level variability.

Table 4 P-values for Birth Weight and Preterm Outcomes Across Variables

Variable	P-value (Birth Weight)	P-value (Preterm)	Test Used (Birth Weight, Preterm)
Parity	4.52×10^{-1}	1.12×10^{-1}	T-test, Chi-squared
Preterm	1.48×10^{-242}	0.00	T-test, Chi-squared
Child sex	6.97×10^{-1}	1.36×10^{-1}	T-test, Chi-squared
Maternal Education	3.00×10^{-1}	1.27×10^{-5}	ANOVA, Chi-squared
Maternal Ethnicity	1.12×10^{-5}	4.18×10^{-4}	T-test, Chi-squared
Area level deprivation	0.00	2.55×10^{-2}	ANOVA, Chi-squared
Maternal smoking	8.91×10^{-2}	1.28×10^{-1}	T-test, Chi-squared
Maternal alcohol consumption	2.51×10^{-2}	3.72×10^{-4}	T-test, Chi-squared
Accessibility to green space	1.13×10^{-2}	6.54×10^{-5}	T-test, Chi-squared
Accessibility to blue space	1.83×10^{-5}	2.18×10^{-18}	T-test, Chi-squared
Cohort	0.00	2.97×10^{-121}	ANOVA, Chi-squared
Maternal Age	7.72×10^{-1}	8.30×10^{-5}	Pearson, T-test
NDVI 100m	3.08×10^{-8}	2.00×10^{-5}	Pearson, T-test
NDVI 300m	9.69×10^{-8}	4.78×10^{-6}	Pearson, T-test
NDVI 500m	1.01×10^{-6}	4.75×10^{-4}	Pearson, T-test
Residential distance to green space	0.00	1.09×10^{-13}	Pearson, T-test
Residential distance to blue space	6.70×10^{-207}	4.19×10^{-304}	Pearson, T-test
NO_2	1.05×10^{-42}	5.68×10^{-1}	Pearson, T-test
$PM_{2.5}$	6.33×10^{-17}	5.49×10^{-29}	Pearson, T-test
Birth Weight	0.00	1.48×10^{-242}	Pearson, T-test

3.4. Predictive Modeling for Birth Weight

To predict birth weight, we initially developed a linear regression model using fixed effects. Statistically significant predictors ($p < 0.05$) were selected based on association tests. Given the strong correlations between variables such as NDVI (100m, 300m, 500m) and pollutants (NO_2 and $PM_{2.5}$), polynomial interactions (e.g., $NO_2 \times PM_{2.5}$) were included. Using best subset selection, the most predictive features were identified:

- Residential distance to green space
- Residential distance to blue space
- NO_2 levels
- Accessibility to green space
- Polynomial interaction: $NO_2 \times PM_{2.5}$

Observations 4441, 4415, 4408, 3511, and 3369, identified as leverage points and outliers, were removed to prevent distortion of model coefficients and ensure more reliable estimates.

3.4.1 Linear Regression Model Results

- Adjusted R^2: the model explained 78% of the variance in birth weight (Adjusted $R^2 = 0.78$), indicating strong explanatory power.
- AIC/BIC: the Akaike Information Criterion (AIC = 52705.84) and Bayesian Information Criterion (BIC = 52742.89) confirmed the model's balance between fit and complexity.

Predictor Coefficients (Fixed Effects):

1. Residential Distance to Green Space: Coefficient = -644.4626 (95% CI: -658.060, -630.865)
2. Residential Distance to Blue Space: Coefficient = -490.9744 (95% CI: -504.831, -477.118)
3. NO_2 Levels: Coefficient = -178.0390 (95% CI: -203.629, -152.449)
4. $NO_2 \times PM_{2.5}$ Interaction: Coefficient = -75.4300 (95% CI: -203.629, -48.868)
5. Accessibility to Green Space: Coefficient = -66.1530 (95% CI: -96.532, -35.774)

3.4.2 Collinearity Assessment

To ensure the robustness of the model, multicollinearity was assessed using the Variance Inflation Factor (VIF). The results indicated no significant multicollinearity among predictors, as all VIF values were below the critical threshold of 5. This confirms that the predictors are independent and can reliably contribute to the model without redundancy. The results are presented in Table 5.

Table 5 VIF values visualization.

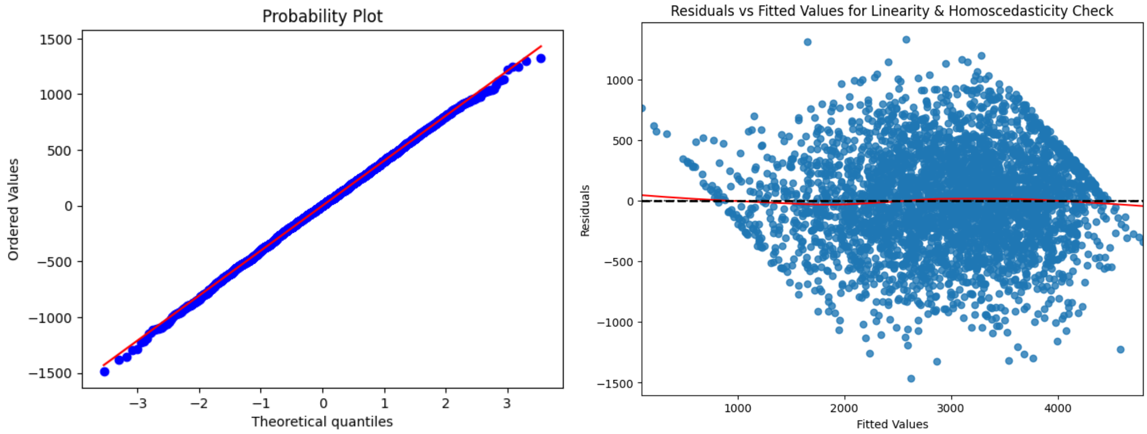
Variable	VIF
Constant	3.85
Residential distance to green space	1.03
Residential distance to blue space	1.09
NO_2	3.64
$NO_2 \times PM_{2.5}$	3.78
Accessibility to green space	1.01

3.4.3 Residual Diagnostics

- Independence: the Durbin-Watson statistic (2.038) indicated no significant autocorrelation.
- Normality: the Shapiro-Wilk test ($p = 0.243$) confirmed normal residual distribution (Figure 2a).
- Homoscedasticity: the Breusch-Pagan test ($p = 0.424$) validated constant variance across residuals (Figure 2b).

These metrics confirm the robustness of the model and ensure reliable result interpretation.

Figure 2 (a) Residual Q-Q Plot – Showing normality of residuals. (b) Residuals vs Fitted Values: Visual Check for Linearity and Constant Variance.

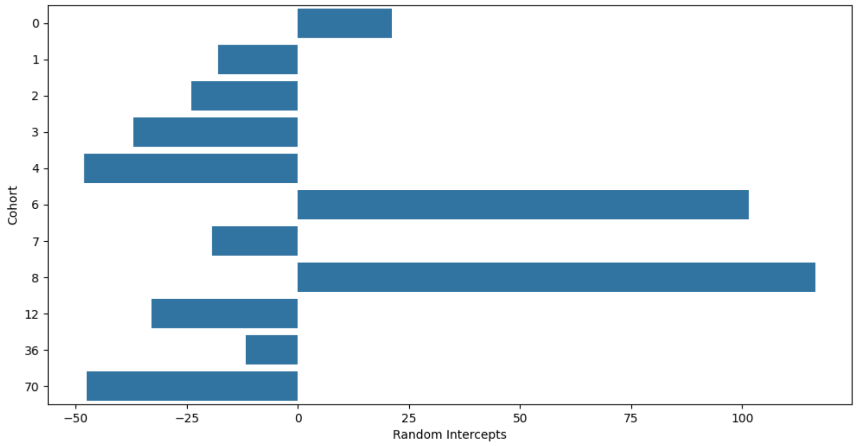


3.4.4 Generalized Linear Mixed Model (GLMM)

To account for cohort-specific variability, we extended the analysis using a GLMM. This approach incorporated both fixed and random effects, providing a more comprehensive understanding of the factors influencing birth weight.

- Fixed Effects: the results were consistent with the linear regression model, reaffirming the influence of predictors such as green/blue spaces, NO_2, $PM_{2.5}$, and accessibility to green space.
- Random Effects (Cohort-Level Variability): cohort deviations ranged from -48.10 (Cohort E) to +116.40 (Cohort H), highlighting local differences in environmental, socioeconomic, or healthcare conditions (Figure 3). These variations emphasize the importance of accounting for cohort-specific effects to accurately model and interpret birth outcomes.

Figure 3 Cohort-specific deviations from the mean birth weight.

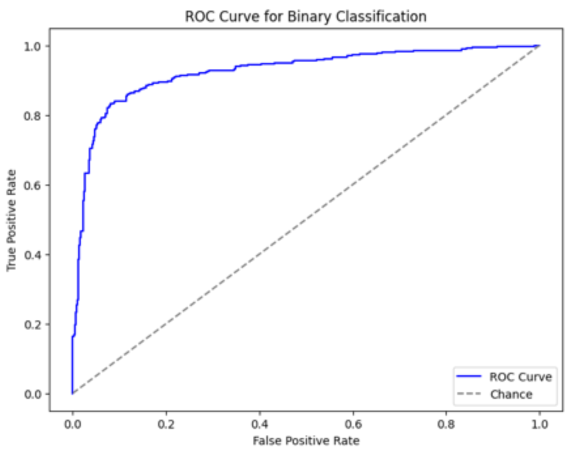


3.5. Predictive Modeling for Preterm Birth

To predict preterm birth, a logistic regression model was developed, focusing on key environmental predictors. Statistically significant predictors ($p < 0.05$) were identified based on association tests. Model selection was guided by the Adjusted R^2. The most predictive features included:

- Residential distance to green space
- Residential distance to blue space
- NO_2 levels
- $PM_{2.5}$ levels
- Accessibility to blue space

Figure 4 ROC curve.



3.5.1 Model Optimization and Performance

The logistic regression model achieved a pseudo R^2 of 0.46, demonstrating a moderate explanatory power for predicting preterm birth outcomes. The model converged successfully after seven iterations, with a log-likelihood value of -1265.4.

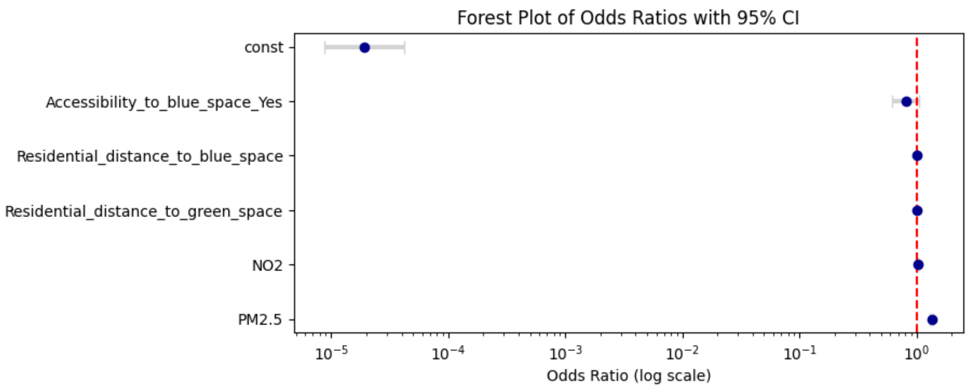
- AIC/BIC: AIC = 2778.85 and BIC = 2815.91 indicate a well-balanced model between fit and complexity.
- ROC-AUC: the ROC-AUC score on the test set was 0.925, highlighting excellent discriminatory ability between preterm and non-preterm cases (Figure 4).

3.5.2 Logistic Regression Results

The odds ratios for each predictor provide insight into their impact on the likelihood of preterm birth (Figure 5):

1. Residential Distance to Green Space: Coefficient: 0.0066. OR: 1.01 (95% CI: 1.01–1.01).
2. Residential Distance to Blue Space: Coefficient: 0.0033. OR: 1.00 (95% CI: 1.00–1.00)
3. NO_2 Levels: Coefficient: 0.0364. OR: 1.04 (95% CI: 1.03–1.05)
4. $PM_{2.5}$ Levels: Coefficient: 0.2961. OR: 1.34 (95% CI: 1.31–1.38)
5. Accessibility to Blue Space: Coefficient: -0.2097. OR: 0.81 (95% CI: 0.62–1.06)

Figure 5 Predictor coefficients with 95% confidence intervals.



3.5.3 Classification Metrics

The model's classification performance was robust, as evidenced by the confusion matrix and derived metrics (Figure 6):

- Accuracy: 87%
- Precision: 91% (Preterm), 85% (No Preterm)
- Recall: 76% (Preterm), 95% (No Preterm)
- F1-Score: 83% (Preterm), 90% (No Preterm)

The high specificity (0.953) underscores the model's strength in correctly identifying non-preterm cases.
Variance Inflation Factor (VIF) analysis confirmed minimal multicollinearity among predictors, with all VIF values below the threshold of 5.

3.5.4 Generalized Linear Mixed Model (GLMM)

The random effects analysis provides crucial insights into cohort-level variability, capturing how differences in local environmental and socioeconomic factors influence preterm birth risk. These deviations, ranging from -0.19 to +0.36, reflect the heterogeneity among cohorts, emphasizing the necessity of tailoring interventions to specific group contexts. Table 6 illustrates these cohort-specific random effects, highlighting the extent of variability across groups.
However, the Generalized Linear Mixed Model demonstrated performance comparable to the fixed-effects model, suggesting that while random effects provide additional interpretability, they may not substantially enhance predictive accuracy in this context.

Figure 6 Confusion matrix heatmap.

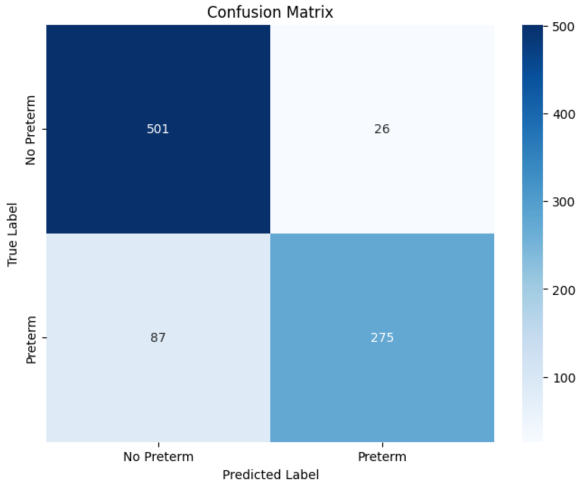


Table 6 Cohort-specific random effects.

Cohort	Random Effect
A	-0.12
B	-0.16
C	-0.19
D	0.06
E	0.10
F	-0.14
G	-0.03
H	-0.04
I	0.13
J	0.36
K	0.04

4. Discussion

This study highlights the significant role of maternal exposure to natural environments and air quality in influencing birth outcomes, such as birth weight and preterm birth. The findings demonstrate that proximity to green and blue spaces is associated with improved pregnancy outcomes. These results align with prior studies suggesting that natural environments contribute to maternal and fetal health through mechanisms like stress reduction, better air quality, and opportunities for physical activity. For instance, shorter distances to green spaces were strongly linked to higher birth weight, corroborating existing evidence on the positive impact of vegetation on pregnancy.

Interestingly, while blue spaces showed benefits for reducing preterm birth risk, their effects were less pronounced for birth weight. This suggests unique pathways through which different types of natural environments influence maternal health, warranting further research to disentangle these effects. On the other hand, exposure to air pollutants such as NO_2 and $PM_{2.5}$ was consistently associated with adverse birth outcomes. These findings emphasize the detrimental impact of poor air quality on fetal development, which supports the implementation of stricter air pollution controls to improve public health outcomes.

While the study found significant associations between environmental factors and birth outcomes, maternal socioeconomic variables such as education level, ethnicity, and area deprivation were not significantly linked to birth weight or preterm birth. This suggests that environmental exposures, particularly proximity to natural spaces and air quality, play a more direct role in shaping these outcomes. However, socioeconomic factors might still exert indirect effects, influencing access to cleaner environments and healthcare resources. Further investigation is needed to clarify these relationships and understand potential mediating pathways.

Cohort-level variability also emerged as a critical factor, highlighting localized disparities in environmental exposures and healthcare access. These variations suggest that addressing community-specific barriers is essential for reducing inequities in birth outcomes.
The findings of this study carry important implications for public health and urban planning policies:

1. Enhancing Access to Natural Spaces: Urban planning should prioritize the integration of accessible green and blue spaces, particularly in densely populated or deprived areas. This could mitigate disparities and promote maternal health.
2. Reducing Air Pollution: Stricter environmental regulations are necessary to decrease exposure to harmful pollutants such as NO_2 and $PM_{2.5}$, which negatively impact fetal growth and the risk of preterm birth.
3. Targeted Community Interventions: Policies should focus on high-risk cohorts, providing localized solutions to address specific environmental and healthcare challenges.

5. Conclusions

This study underscores the critical role of natural environments and air quality in shaping pregnancy outcomes. Proximity to green and blue spaces is associated with improved birth outcomes, while exposure to air pollution exacerbates risks. Addressing these factors through urban planning and public health interventions can lead to significant improvements in maternal and fetal health. Moreover, acknowledging cohort-level disparities is vital for designing equitable and effective policies. These findings provide a roadmap for enhancing health equity and promoting better birth outcomes through informed environmental and healthcare strategies.

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